

Mia Consalvo

Cyber-Slaying Media Fans: Code, Digital Poaching, and Corporate Control of the Internet

This article examines media fans' Web sites devoted to Star Trek and Buffy the Vampire Slayer to determine how the code of the Internet allows fans some creative control over media artifacts of interest to them and how such actions pose a challenge to media corporations and their copyrights. However, this challenge is also limited by the changing code of the Internet, at varying levels, as corporations seek to change open code to closed-proprietary code and limit such actions. Thus, this article explores the resistances of fans to corporate actions and how the code of the Internet is gradually changing into a system more conducive to corporate capitalism than community creation or any other transformative space.

It would be unwise to ignore the design of cyberspace itself while we are engaged in the myriad considerations of particular GUI [graphical user interface] and VR [virtual reality] implementations.

—Benedikt (1992, 131)

Cyberspace is architecture; cyberspace has an architecture; and cyberspace contains architecture.

—Novak (1992, 226)

Contributors to the volume *Cyberspace: First Steps* wrote of the radical potentials for cyberspace, mostly visualized as the development of virtual reality (VR) systems where users could interact socially as well as professionally (Benedikt 1992). Writers in that volume were concerned with the architecture of cyberspace, as they believed that how the digital space was built would encourage and discourage certain uses. They knew that architecture could never completely determine user actions but was a critical dimension to consider. However, VR's potential as a popular medium has largely faded, and most attention to "cyberspace" now concerns the World Wide Web and com-

mercial activity. Yet the arguments of Benedikt and others concerning architecture are still important in considering that “new” space. With that in mind, this article considers how mainstream users of the Internet, and more particularly the World Wide Web, use the architecture of that space for their own purposes and how that architecture constrains their activities in unseen ways.

The Web was focused on because of its popularity as well as the opportunities it still offers amateur users for personal creation. Here, its architecture is explored as “open” and “closed” code. Ideologies are constituted within the various layers or levels of code that make up the Web, and changes in code can change ideologies. How the architecture of the Web and HTML code in particular allows users to challenge society’s increasing “grid of control” and how that changing architecture also works to limit or erase that potential are explored in detail. To do so, the activities of media fans are examined. Fans of television shows such as *Buffy the Vampire Slayer* and *Star Trek* are active online but not in overtly “radical” or “subversive” ways. They provide telling examples and evidence of how that new media environment is changing and how users both challenge and are limited by increasing control of the Web. The architecture of the Web allows fans some “resistant” activities but, at the same time, works to constrain these activities in important ways.

Media fans are good candidates for this study, especially as their activities on the Web are not overtly “political.” What they can and cannot do in this society and on the Web are important indicators of how Internet code is changing and what can be expected in the future regarding corporate control and fair use of information or intellectual property.

Why Study Fans?

One way researchers have investigated meaning making by individuals is through the study of media fans. These fans are usually specific to a television show, movie, or genre. Early research disputed the notion that fans were unquestioning “groupies” of media, who eagerly (but unquestioningly) accepted the texts they viewed. Researchers found that fans regularly take an active approach to interpreting media texts and can be quite critical of the shows they say they love (Baym 1995; Jenkins 1992). They were often found to be sophisticated in their critiques as well as praises. Fans are ideal active viewers who consume but also question what they see.

Another reason to study media fans is their creation of supplementary materials in relation to the primary text of interest. Bacon-Smith (1992), Jenkins (1992), and Lauters (2001) have found that many fans create such things as fan fiction (fanfic), poems and songs about favorite characters and stories, scripts for further episodes, and even awards for the best of these fan creations. Thus,

many fans are not only active in their understandings of media texts, but they take their activity further to the creation of related materials.

Early media fans often congregated around annual conventions to meet and share their work, but with the development of the Internet, a new way for sharing information, chat, and creation of fan-made products was found and quickly seized upon. Early use of the Internet by fans was often on newsgroups and listservs, where fans discussed episodes of television shows, shared fanfic, and reviewed shows (Baym 2000; Consalvo 1995). Scholars such as Susan Clerc (1996) have discussed how gender dynamics among fans began to alter through Internet use, while others have examined how the Internet has added to the fan experience in various ways, including the creation of a larger community, more opportunities to share creative activities, and new ways to recognize these creative endeavors (Baym 2000; Lauters 2001). Fan activity on the Internet allows for greater creative expression and, from its beginnings, has included a wider demographic of Internet users (the majority of media fans are women) than the original “middle-class white male” user that was initially defined (Consalvo 1995).

Finally, fans are important as new media users because their activities provide a challenge to media corporations concerning intellectual property and its limits. Fans understand that they are “poaching” materials owned by others (Jenkins 1992), but their actions and creations call into question the limits of ownership when intellectual property becomes part of a common culture. Texts that hold shared meanings for a significant number of people become something more than the “property” of a corporation. And how corporations react to that growing influence and poaching is important to investigate.

For all of these reasons, I chose to study fans’ use of the World Wide Web because fans provide a somewhat cohesive group, often commenting on others’ Web sites, connecting their sites, and actively creating multiple sites on similar themes. In these activities, they are the ideal active users of the Internet, and they are also distinct (in design and content) from commercial sites online. I chose fans of *Star Trek* and *Buffy the Vampire Slayer* for several reasons. Fans of *Star Trek* have a long history of active participation in commenting on and contributing “new” material to the series and movies. The one-hour drama *Buffy the Vampire Slayer* began airing in 1997, providing a show with less history, yet equally enthusiastic fans.¹

What makes these fans salient is that they are not passive in response to corporate actions they deem threatening to their activities. Fans of both shows have quickly responded to corporate threats and have done so very publicly. For example, the owner of *Buffy*, Twentieth Century Fox, cracked down on fan Web sites just as Viacom did with *Star Trek* fan sites. Fan responses to both actions were swift.² For these reasons, fans of both *Buffy the Vampire Slayer*

and *Star Trek: Voyager* were chosen, and their Web sites were the primary documents examined.

Fan Web sites centering on *Star Trek* and *Buffy* were identified, downloaded, and analyzed. The Web sites chosen for this analysis are typical of online media fan offerings. Yet this article does not claim that it has conducted an exhaustive search or definitive analysis of all fan Web pages devoted to *Star Trek* or *Buffy*. This is because it is currently practically impossible to identify and then study every Web page devoted to a particular topic, and more important, exhaustiveness is not the goal of this project—rather, an understanding of some of the more common features and uses of the World Wide Web is sought. This article therefore takes as its starting point two groups of media fans and their Web sites.

Users as “Cyborgs”

In addition to the narrowed focus from “users” of new media to “fans,” it is also useful to explore how these users are becoming cyborgs in a metaphorical sense. Perhaps the best known metaphorical cyborg was the model proposed by feminist scholar Donna Haraway (1991), which disrupts artificial dualisms in human thinking. Because the cyborg is neither all machine nor all human, it challenges the purity of both categories. Haraway believed feminists should embrace the concept of the cyborg as it calls into question other “artificial” dualisms, including male/female, human/animal, and culture/nature. Furthermore, Haraway argued that we should embrace technology and its role in our future, as to do otherwise is to cede control of that potentially destructive force to dominant, conservative forces already in charge.

Other theorists have taken the concept of the metaphorical cyborg further, looking at how humans and communications technology are becoming intertwined—particularly with the developments of the computer and the Internet. Sherry Turkle (1995) brought humans and the Internet together by suggesting that technologies and how we use them are changing our selves as well as our bodies. She wrote,

As human beings become increasingly intertwined with the technology and with each other via the technology, old distinctions between what is specifically human and specifically technological become more complex. . . . Our new technologically enmeshed relationships oblige us to ask to what extent we ourselves have become cyborgs, transgressive mixtures of biology, technology, and code. (P. 21)

Alluquere Stone (1996) made a similar argument—because we are all so reliant on electronic communications technologies to live our lives, we have

indeed become cyborgs. Our enmeshment with that technology has altered us, with unknown results. Turkle's and Stone's theses are intriguing and suggest that with the growth of the information economy, we are seeing the dawning of a shift from print culture to an information/cyborg culture.

Our reliance on the information system continues to grow. While we may not "need" to log on every day and surf the Web, the system that produced the Web is necessary for the daily operations of our computerized infrastructure of government and business. The Y2K problem was a good case in point. One simple problem—computers switching from the year 1999 to 2000—threatened banks, air traffic control, and telecommunications systems. Stone is correct in her thesis that as we rely on these communications technologies, we become cyborgs. We just do not fully realize it yet. Our forays into chat and interaction are the tip of that iceberg, although they signal our increasing reliance and ultimate complicity with these systems of control. Each time we surf the Web or read our e-mail, we give tacit approval for the perpetuation of that system. As we invest more time and attention to the system, it grows more real, more accepted. As we take on its terminology, integrate it into our daily lives, it becomes a part of us. While that active move by us is currently performed only by the privileged few, the underlying system is falling into place for all. Computers house video store rental choices, medical records, and automatic teller machine transactions. As that information is increasingly linked and turned into a commodity, we are all watched and assessed, much like in Foucault's (1975) panopticon.

So, it is important to consider how users of new media, here specified as fan users, are cyborgs. They are cyborgs as their dependence on communications technologies grows but also in more radical ways, as Haraway proposed. She argued that it was only as cyborgs that we could challenge that control and reshape (or alter) new media technologies. That challenge comes from resistance to dominant ideologies, as the process of hegemony is worked out in daily practices (Abercrombie 1990). For example, fans' use of copyrighted materials constitutes a challenge to corporate claims of complete ownership of intellectual property. And how corporations respond to such challenges demonstrates how the hegemonic process both incorporates some "oppositional" views as well as blocks others—all to stay in dominance. The potentials and limitations of that resistance and counterresistance, and how they are expressed by *Buffy* and *Star Trek* fans online, are explored in this article.

Fan/Borgs' Deployment of Code

Fans as cyborgs (or fan/borgs as I abbreviate it here) use the Internet and the World Wide Web in particular for their creations, activities, and communication. For the fan/borgs who venture beyond e-mail or posting messages on Web

pages, that will include the coding of HTML. The deployment of HTML can vary in sophistication—from the individual who uses WYSIWYG³ editors such as Microsoft FrontPage to the more adventurous person who codes in “raw” HTML, using only Simpletext as his or her editor. Yet all, to different degrees, learn HTML code to create Web pages, update them, and add media artifacts of interest. And for all of them, code plays an integral role in what they can do (both legally and illegally), how they can do it, and how other fans receive that information.

Lessig (1999) wrote that code is indeed important as it is the architecture that structures the Internet. One of Lessig’s central arguments is that the rhetoric of the Internet as “naturally free” from regulation and control is false. Rather, the Internet has been built through code, and specific decisions and values all went into that architecture. The earliest programmers may have created a system that allowed for anonymous interaction, or difficulty in tracking the “true” identities of users, but that result is not an absolute. To talk about the “nature” of the Internet is to give agency to a medium, rather than its creators. So in reference to fan/borgs, to suggest that their actions are free from regulation or control, or the Internet is a place for them to act without any penalty, is false.

A further distinction must also be made when talking about fan/borg activities on the Web. The architecture or code that composes the Internet is not simply one code but several levels of code. The code governing the Web is one of these, while other codes govern interactions between servers and operating systems and other types of connections. So, it is not possible or reasonable to talk about the code of the Internet, as there are many levels of code present. And in talking about fan/borg use of code in relation to the Web, it is only at the level of HTML coding that fan/borgs work, leaving other levels to developers and scientists.

Fan/borgs’ use of HTML code is important to study because it is the central level of code open to use by nonprogrammer Web users and is the layer that most Internet and Web users see and experience as “the Internet.”⁴ As Lessig (1999) argued, code carries ideology, and HTML has embedded within it ideologies about expression, use, copying, and replicability. It is relatively open, as most browsers will let a Web page viewer see the “source code” of the page and determine how it is put together. This code can also be copied and pasted to a new page, allowing users to take code they like and apply it to their own creations. Furthermore, browsers showing these Web pages also allow “drag and drop” capabilities, meaning fan/borgs can find photos they like on a Web page and simply move them onto their local machines—for easy reposting on their own sites. Thus, fan/borgs’ use of HTML code can challenge established commercial practices and laws surrounding them. This resistance to dominant

business practices is more blatant than past fan practices, and that use is embedded within one level of the current coded technology.

Technology Is Not Neutral

The idea that uses and values are coded into technologies is part of a larger argument that technology is not a neutral or passive tool. Carey (1989) and others have argued that technologies must be studied as dynamic processes that are developed with uses and values embedded within them but that also can have these uses and values altered through use.

Ellen Ullman (1995) wrote that the work of creating computers and the code to run them is defined by a specific group of people—male engineers—who have their own peculiar culture and practices. This results in “the engineer’s assumptions and presumptions” being built into code (p. 143). Ullman also believed that this particular culture is very different from the mainstream, which could lead to troubling consequences as “computer culture” starts to take over more aspects of our lives.⁵

Originally, developers of the Internet created hardware and software that were not proprietary and thus open to sharing, copying, and tinkering with. Most of the Internet was developed by a relatively homogeneous group of people, and the values of the group stressed openness and sharing in their work, to develop the technology as quickly and efficiently as possible (Rheingold 1993).

But as Lessig (1999) explained, code for software and hardware could not be completely individualized, otherwise communication would be impossible. With only general standards to adhere to, and no group yet competing for proprietary ownership of code, the architecture of the Internet (and World Wide Web) was at first relatively open and free. In addition, a majority of work on the Internet began in the United States, and so national ideologies were likely to find their way into the code and technology. Thus, ideas about individual rights to privacy in communication and beliefs in progress through technological development were embedded in the developing Internet, as well as in discourse about it (Rheingold 1993).

And that process of developing ideological technologies happened multiple times, on multiple levels, for the Internet includes such things as Internet browsers, LAN systems, and transmission control protocol/Internet protocol (TCP/IP). Although all of that code needs to have enough in common to work together, minor differences may occur, and each level has its own meanings and functions. All of these ideological technologies come into play as fan/borgs, active users of the new media, seek to use that technological medium to advance their own goals of sharing and creating materials of interest to them

and their (loosely defined) community. So how does that play out in terms of the ability to manipulate code?

Top Level

Fan Use of the Internet and HTML Code Creation

Fans can “deploy” code on the surface level of the Web, meaning that they can create, alter, modify, or use code in different ways. In contrast, fans use other parts of the Internet, including Web browsers and connection software, but are unable to significantly modify or alter these parts and usually engage them in a “read-only” mode. That is an important distinction—HTML coding is possible but altering the code of Internet Explorer is not. Allowable HTML coding involves Web page creation, including hypertext links; the creation and insertion of text, images, video, and audio files; and design of Web pages. Also included is the ability to view other sites and examine (and borrow) source code, as well as copy text or other items to the user’s computer for later posting to one’s own Web site.

Fan/borgs can do all that because HTML is relatively open—it allows users of the code to become creators fairly easily. Fan/borgs manipulate that HTML code for more and less sophisticated creations. For example, online fans create artwork, stories, compilations of hard-to-find pictures, audio files, and video clips, as well as summarize and review past episodes. Many of these activities are not native to the Internet—Camille Bacon-Smith (1992) has documented how female fans of *Star Trek* created fanfic long before the advent of the Internet. However, the Internet has become a preeminent publishing opportunity for media fans as it guarantees an international audience, as well as greater access to materials suitable for “borrowing.” But the most important deployments of code by fan/borgs for this article are use of hypertext links and Web rings.

In addition to their own creative endeavors, fans almost always provide links to other fan sites that focus on *Star Trek* or *Buffy*. This is done through the listing of specific sites, as well as through the inclusion of Web rings. Web rings are not sites themselves but networks of related sites. Web rings for *Star Trek* include “The Borg Collective Web Ring,” “The Great Link,” and “Locutus Assimilation Ring.” Web rings for *Buffy the Vampire Slayer* include “Sonja Marie’s Buffy the Vampire Slayer Links Webring,” “The Buffy Ring,” and “Slayer Pride Webring.”

These Web rings allow the interested user to either jump to another random site in the ring or the next in the “chain.” By making that connection, site cre-

ators show interest in one another's sites and support access to all sites. Fans also claim connections with other fan Web sites, sites often too numerous for the individual fan to have identified and listed on his or her particular site. For example, "The Buffy Ring" boasts "over 350 Web sites," and the Locutus Assimilation Ring claims to have more than 1,000 members (retrieved 1 November 1998, from <http://www.geocities.com/Area51/Vault/7279/Ring.HTML>). In coding these links, fans are acknowledging the interconnectedness of their projects, as they seek to bring their sites together for others and themselves—joining them together into a larger statement or belief about *Star Trek* or *Buffy the Vampire Slayer*.⁶

Thus, through the use of the link, the fans are creating unity, if not specifically a community, where previously there was none. This is achieved not through sites of particular interest themselves but instead networks of self-organizing sites. The ring represents not a destination but a relationship. This hypertext coding decentralizes authority at a certain level, as a function of the Web architecture encourages users to "join up" with others and become a collectivity. A technological function is here exploited by fan/borgs to create a group based on technology, giving those in the group affiliation with a larger "cyborgian" entity.

In that regard, fans are actively creating and sharing texts, and in the process, the relationship of the (fan) reader to the "original" text has changed. A fan's deployment of hypertext links on his or her Web site calls into question the belief that one reading—even the official one—is preferred. Instead, multiple "readings" of *Buffy* or *Star Trek* become the norm. Furthermore, the mixing of the official and unofficial occurs when fans link official sites to their own, and also when fans "borrow" official materials and offer their own views in response. Jenkins (1992) would argue these fans are engaging in poaching of cultural artifacts—reappropriating them for their own purposes. But in addition to engaging in that poaching, fans, by virtue of the medium they have chosen for that deployment—hypertext—are also engaging in a decentering and recentering of media texts. George Landow (1992, 11) wrote that the ability to link disparate sites or texts destroys not only the idea of the linear but also the idea of the center. The "original" center is the authorized story plotlines and characters created by corporations. However, with the ability to connect official and unofficial sites, fans have begun a move that decenters official, received ideas about media texts and recenters ideas on their own versions of the stories.

For example, fans of *Buffy* can go beyond the television series to explore other plotlines and character developments they find interesting. Sonja Marie's "Slayer Links" are Buffy-related sites in a general Web ring—yet these links also include sites for fanfic. This list claims 988 different sites or

stories (Sonja Marie's "Buffy Fanfiction Links" 1 August 2000, <http://www.geocities.com/TelevisionCity/7728/fic.html>). These stories vary in length, quality, tone, and subject matter. Most stories explore relationships between different characters or create new plots. For example, some stories keep the relationship between Buffy and Angel going, while others posit a lesbian relationship between Buffy and another slayer, Faith. Fans of *Star Trek* have long done the same thing. Indeed, creating gay story lines for characters had its beginnings in the "slash" fiction produced by *Star Trek* fans that wrote of a relationship between Captain Kirk and Spock. These actions acknowledge the "official" characters and do not claim overt ownership of them, yet in that appropriation, fans are challenging the creators' view that ownership is strictly held by a corporation. Certainly, the creative material is the product of fans, and that blending of ownership, encouraged by HTML code, suggests that what is central to a particular media text is becoming more and more slippery, and more difficult to control. As Tollander (1992) explained in relation to cyberspace generally, "the legal definition of ownership is problematic in collaborative cyberspace" (p. 323). So, the move from physical to digital space challenges traditional ideas of property and ownership—ideas that fans had always contested but that find new urgency on the Web.

Fans' code deployments demonstrate resistance or a challenge to dominant ideologies concerning copyright laws and illustrate that fans are active readers and creators, moving in and through different types of reading as well as centering at different times. This more active role must be accounted for in the examination of not just *Star Trek* and *Buffy* but of popular culture in general, as what is viewed as popular culture can now be recentered by fan activity into not a static, received version but a modified version put forward through fans' creative actions.

However, while fans may be reorienting some of their online activities away from official discourses, they would not think that their creations are the center of their primary mediated experience—that remains viewing the shows. While some fans argue that certain episodes or seasons are not true "canon" (meaning that the plot or character actions stray from what the original creators set the standards for), the center of their activity does remain (for most fans) the corporate-produced media product *Buffy* or *Star Trek*. This occurs unless the text in question has finished its official run, as was the case with *Lois and Clark: The New Adventures of Superman*, where fans began writing new seasons for the show (Lauters 2001). But even if fans do not claim to challenge the official production's place at the center, the production company does not automatically hold a place at the center of all fan experiences in regards to a particular text.

Fan/borg activities can thus provide challenges to traditional notions of intellectual property by suggesting that although media companies are respon-

sible for the primary text, their control should not extend over all uses of that text. Fan/borgs thus take some control from media corporations, and their activity is possible through the openness of HTML code.

In sum, hypertext and linking (as deployments of HTML code) make the center of fan activity a “moving target” where previously the center was more fixed—it used to be the official representation as put forth by media creators. While fan activity is a valid form of expression acknowledged by creators, it has generally been considered a peripheral activity by those same creators. But with the increased use and potential for manipulation of hypertext and HTML code, that is changing. How the media corporations react to that poaching and decentering further illustrates deployments of code as well as recourse to the hegemonic process.

Corporate Reactions to Digital Poaching

Media companies are concerned about the implications of fan-coding activities online and are taking action against what they perceive as infringement of copyright laws. Both Viacom-Paramount and Twentieth Century Fox have objected to fans’ uncontrolled use of various media texts, demonstrating how important the center is and the lengths they will go to maintain it. Their actions also demonstrate different approaches to compensating for the openness of HTML code, through either the marginalization of (or punishing of) oppositional activity or the incorporation of some activities into the hegemonic, offering some fan activities limited legitimation.

Paramount Pictures and its parent company, Viacom, have been very aggressive in pursuing certain fan Web sites they believe are unlawfully using copyrighted *Star Trek* material. They have threatened many site creators with lawsuits, and in response, some sites have removed offending materials while others have simply shut down. These legal actions taken by Paramount-Viacom demonstrate their awareness of the recentring of material about *Star Trek*, and they fear fan sites are becoming more popular than the official sites. The technological recentring by fan/borgs potentially pushes Paramount out of the dominant center, where they have always been (while almost all fan Web sites have links to other sites, Paramount’s official page does not list any links to unofficial sites). In response, creators such as Luca Sambucci have been ordered to take down certain materials from their sites due to copyright infringement.

Likewise, Twentieth Century Fox television, which owns *Buffy the Vampire Slayer* and previously sold it to the WB Network,⁷ also warned “several well-trafficked fan Websites” that they needed to remove copyrighted material or face a lawsuit (Keller 1999). In that regard, both corporations responded to the “threat” of code deployment in a traditional way—the legal system.

However, the WB Network (while it still aired *Buffy*) also took a different step—opting to encourage new norms of use. In that regard, the corporation attempted to incorporate the resistance by allowing previously “illegal” activities to continue—but only under their terms. Thus, the hegemonic system allows some changes but leaves core power relations unaltered. The WB Network developed a space—AcmeCity—for fans to create *Buffy* pages without harassment. Fans were given 20 megabytes of free space and free photos of the *Buffy* cast to create these sites. The WB created a space for fans that was not antagonistic and acknowledged the importance of fan attention to their media products. Yet the site AcmeCity was not mentioned on the (original) official *Buffy* site. How exactly fans were supposed to find out about and use AcmeCity was (and now remains) unclear. Here, the WB (and by extension Twentieth Century Fox) acknowledged some loss of centrality but attempted to limit that loss by keeping the fans (and their pages) at arm’s length, still slightly peripheral to the alleged “center” of *Buffy*. This alternative response also tried to “correct” certain uses of code by encouraging alternate code creation. So the new “social norm” being pursued, as Lessig would argue, attempted to “let” fans deploy code but only in limited ways.

Fan Reactions to the Threats

In response to the initial threats of lawsuits, fans were not passive. While some dutifully removed contested materials, many refused to comply and also challenged these actions through the further deployment of HTML code. This included a “blackout day” for *Buffy* fans and the development of an “Online Freedom Federation” for *Star Trek* fans. The Web site for the Online Freedom Federation served as a site where like-minded fans could sign a petition to protest corporate actions and view reports on press attention, as well as fan reactions to the ongoing activities. Fans of *Buffy* called for “Operation Blackout” on 13 May 2000, when their Web sites went offline, “in hopes of emphasizing the importance of free publicity the sites generate for Fox” (Keller 1999). This response, in addition to a more traditional letter-writing campaign, demonstrated that fans will use any means (and code) at their disposal to assert their interests. In that instance, fans used the technology as a tool for further resistance, demonstrating how as fan/borgs they adapted to the technology and used it very adeptly.

In addition, HTML coding was deployed in a fascinating way by one particular *Star Trek* fan. This fan deployed a political use of the hypertext link to disinclude or actively block access to specific Web sites. This occurred in relation to Paramount’s *Star Trek* sites. Luca Sambucci’s Web site, “Star Trek: WWW,” was served with a “cease and desist” letter from Viacom, allegedly due to perceived copyright infringements. In response, he has kept his links to

Paramount and other corporations associated with *Star Trek* present on his pages but deactivated them and stated near the link(s), “We have decided to deactivate this link because of Viacom’s recent actions against the Star Trek community. We will restore the link as soon as Viacom accepts to find a solution [*sic*] together with the Online Freedom Federation” (Sambucci 1998).

Fans have always appropriated copyrighted materials for their own uses (mostly but not all noncommercial), but current fan creations are often more blatant than in the past. Yet just as that new technology lets fans creatively “borrow” material, it also gives them a (limited) form of protest against those who would seek to limit this usage. This creative co-optation demonstrates that fans feel media content is just as much theirs as it is Paramount’s or Twentieth Century Fox’s, and the Web has become a new site for that battle of appropriation and reappropriation (as well as the decentering and recentering discussed earlier). As the deactivated links and the blackout day suggest, fans are exploiting their cyborg status, deploying new and creative ways to protest what they see as unfair practices, and actively exploring the limits of the Web and HTML code to achieve their goals. In doing so, they are questioning the rules of the old system of copyright and ownership as well as creating new uses for developing technologies. By deactivating a link and stating the reasons for doing so publicly, rather than simply removing the link, a fan has attempted to make it harder to access Paramount’s Web pages, but more important, the code deployment symbolically de-links or de-associates the pages and the fan from that other entity—the corporation. George Landow (1992) argued that

hypertext’s very emphasis on connectivity means that excluding any particular bit of text from the metatext places it comparatively much further from sight than would be the case in print technology. . . . Unconnected texts are experienced as lying much farther off and availability and accessibility become very different matters. (P. 187)

In Sambucci’s move, Paramount is left out of the ring, out of the unified group, out of the club. Paramount’s official site, “The Continuum” (retrieved 25 June 1998, from <http://www.startrek.com>), is not only symbolically out of sight, it is literally in this case out of site. The link, the symbol of inclusion, has become a symbol of a broken promise, a part of the whole left disconnected and separated from the rest. Likewise, Operation Blackout demonstrates how integral fan Web pages are becoming in the consumption of popular culture and also points to new ways for fans to protest and make their voices heard.

So while the act of deactivating a link or blacking out a site may be a small protest, it points to the larger significance of hypertext (and HTML) as an element of open code, which allows individuals some abilities to “talk back” to media conglomerates on a relatively equal footing. These deployments and resistances by fan/borgs occur at a “high” level of code found in Web/Internet

use. This code is also relatively easy to learn, in part through its continued openness. Thus, current HTML code allows or gives fan/borgs abilities to challenge media corporations about the “correct” usage of media texts. Furthermore, in acts of poaching, borrowing, altering, and perhaps even “stealing,” fan/borgs also call into question their acceptance of current legal notions such as ownership of intellectual property and its fair use.

Just as fan/borgs seek to contest media corporations’ ownership and control of media texts, those same corporations recognize the fine line of fan activity and seek some sort of balance in maintaining fan interest while protecting “their” properties. Corporations’ attempts at sanctioning fan Web sites, or encouraging fans to create “acceptable” sites, indicate they know that activity at that level of Web use is contested, with each “side” having certain tools at their disposal. Fans’ reactions to corporate actions show that fan/borgs can be innovative deployers of HTML code to protect their interests. And corporate reactions to fan Web activities demonstrate how these businesses are struggling over the implications of open code. As Lessig would argue, corporate responses to fan/borg uses of code (legal, social norms) represent new ways of trying to “control” these practices and this space—indicating the importance of the activity. The hegemonic process here works to contain certain activities, but others are still rejected. However, the lack of a complete “victory” over fan/borgs indicates that this level is still in flux, at least as the code (and hegemony) presently stands.

Lower Levels

What Is beneath the Surface?

HTML is only a surface layer of code in regard to the Internet. Just as this article has explored the various uses or deployments of HTML, it is also time to consider other aspects of the Internet to see how HTML functions in relation to the complex web of programs of which it is only one part.

HTML is accessible in the sense that many users who do not have knowledge of computer programming can learn the language or a program to create it and become fairly proficient in its use. Although HTML is changing daily, and very sophisticated programs such as Flash and Fireworks are increasing the learning curve for users, the code still allows almost anyone to design their own home pages and have a presence on the Web. This accessibility is not the case for other types of code.

As Lessig (1999) explained, HTML code can only “run” because of deeper levels of architecture that govern how information exchange occurs on the Internet. These lower levels handle such things as network interactions and

data routing (p. 101). These levels are “the essential plumbing of the Internet, hidden in the Net’s walls” (p. 101). These levels have their own ideologies, such as the lack of unique identifiers for transmission of data, allowing data to travel anonymously and ensuring privacy for senders and receivers. However, these levels are far more uniform in expression than higher levels. That is because the deeper the layers go, the more similarity is needed for the exchange of data. So, there are agreed upon protocols that do not and cannot change, allowing the system to function. At these lower levels, my concern is not over uniformity but about open versus “closed-proprietary” code and the growing dominance of the latter.

HTML code is understood as “accessible” to the nonprogrammer, meaning that almost anyone could talk about how to improve the code. However, lower layers are discursively constructed as not accessible—or interesting—to the nonprogrammer. Crucially, those that work at the lower levels are constructed as experts who know best how these levels are run and so are in the best position to decide the structure of the Internet. However, that leaves the nonprofessional with little or no voice in determining such future directions. While fan/borgs and low-level information technology workers do have some knowledge of HTML code, that is not constructed as professional knowledge and so does not allow them entry into the conversation.

For example, Dorer (forthcoming) found that female information technology professionals in Europe who were paid for Web page construction did not consider themselves to be “real” programmers. One respondent referred to what she did as “child’s play” compared to other types of programming. Dorer’s observations suggest that HTML coding is perceived as less technical and therefore more accessible to the “average” individual. While it is certain that specialized bodies of knowledge are necessary to program at lower levels such as TCP/IP, that “commonsense” positioning of HTML code for the lay person and TCP/IP for the “real programmers” reifies a division in what is allowable change by citizens and what should be left to the “professionals.” Hegemony therefore functions to keep access (and interest) in deeper levels restricted to professionals.

Similarly, Ellen Ullman (1995) wrote that software programmers themselves draw distinctions in regards to the levels of coding that they engage in for their work. She explained that the “lower” the code, the more prestigious the work is considered. Programming software that will be used by consumers is considered the worst of all possible jobs. These programmers are encouraging the attitudes of Dorer’s respondents, to consider HTML coding inferior, almost menial work. Thus, the lowest levels of IPs or software should only be handled by the “true” professionals, and all others should leave that programming to them.

And the lower levels ultimately determine how it all works and what future directions the Internet will take. These future directions are determined by a select few professionals. And who are these professionals? Initially, they were scientists, paid to create a system that was noncommercial and nonproprietary. And so this development community shared its protocols and advancements to create an "open code" system as quickly as possible.

However, the increasing privatization of the Internet has led to the "closing" of these levels, as code becomes more proprietary in its development and use. Lessig (1999) explained that now, code is mostly written within companies such as Microsoft "where standards are the product of competition; where standards tied to a dominant standard have advantages" (p. 207). And

with closed code, users cannot easily modify the control that the code comes packaged with. Hackers and very sophisticated programmers may be able to do so, but most users would not know which parts were required and which parts were not. . . . Closed code is the propagandist's best strategy—not a separate chapter that the user can ignore, but a persistent and unrecognized influence that tilts the story in the direction the propagandist wants. (P. 107)

Thus, while the fan/borg can deploy HTML code to challenge some architectures of control on the Web, the deeper levels of code can now work against these efforts, unless they happen to be hackers or programmers themselves. The corporations now in control of these levels are able to modify HTML in ways more acceptable to the corporations and not the fan/borgs. For example, corporations have created new multimedia formats that circumvent the easy "copy and paste" usability of older standards. So, if UPN posts clips of *Buffy* episodes as Macromedia Flash movies, there is no way for fans to "take" them as the code for that digital format prevents downloading. Here, the resistance of the cyborg is limited, and hegemony reasserts dominant ideologies through discourses of professionalism and the development of new business products that do the (invisible) work of blocking resistance, silently and effectively.

In addition, as the lower levels are constructed both technically and in popular opinion for "professionals only" and are unnoticed in day-to-day operations, the work of the lower levels fades from view, as do their ideological functions. So, fan/borgs may have a blackout day for *Buffy* on their Web pages, but they do not consider how LAN protocols might reinforce the position of the corporation they challenge. Who questions plumbing but the plumbers?

So, while fan/borgs can and do exercise some control over HTML code and use it to challenge media corporations, their actions do not extend to the lower levels of code. Because fan/borgs are not (generally) professional programmers, their knowledge of the deeper levels is limited, and current ideologies about programming encourage them to think that way. Thus, lower levels remain mostly "off-limits" for all but professionals and curious hackers. And

at these deeper levels, code is changing—it is closing in design, with the result that corporations have greater control at these levels, which ultimately affects the surface. And without the skills or the interest in challenging control at these levels, fan/borgs are unable to fully challenge media corporations about ownership and fair use of media texts. Resistance is not futile—it is absent.

Conclusions

Looking back at scholarly attention to the Internet a decade ago, it is amazing and discouraging to see how that communications technology has developed. The earliest predictions about the Internet suggested that users could employ it to revolutionize communication, government, and community and inhabit “virtual worlds” where we could socialize and work with others in ways that would overcome localized prejudices and problems.

Part of the problem with the early predictions was that they were not contextualized fully, and in the rush to proclaim bold new uses, academics as well as industry “true believers” overlooked how past communications technologies have developed and how ownership as well as use play a large part in determining outcomes. The privatization of the Internet in the United States has helped determine its future directions. While early rhetoric included the possibility of “public spaces” as well as universal service, such talk has dimmed, and it is now expected that the Internet is mostly commercial space. Accompanying this development is a “hands-off” atmosphere by government regulators, suggesting that the current trend toward concentration of media ownership will extend to the Internet.

These fundamental changes in the governance and control of the Internet have led to changes in the surface and deep structures of the Internet. What began as a simple protocol for the sharing of research texts (HTML) has become a platform for the transfer and viewing of sophisticated graphics, multimedia, and animation. These developments have consequences for those interested in participating in this world.

The Internet is still dominated by the developed world, and in the West, it is becoming a standard for communication. But even among the privileged in the West, how many actually know HTML? Even smaller still is the group that knows the lower levels of programming code. This group is an elite minority—now mostly privatized and proprietary.

Even if we are living in an information/cyborg culture, those that have access to the tools for creating the culture are quite small. Fan/borgs are challenging some aspects of code use and ownership of media products. They have called into question the notion of a stable center for “ownership” of ideas or culture, and they have worked to create more diversity in understanding media content. Their resistance to dominant ideologies of corporate control of media

content shows that technological tools can be used to challenge grids of control in addition to building them. Their actions are important enough that corporations feel the need to respond.

Yet their actions, as privileged as they are, are only at the surface level of coding for the Web. They can manipulate and deploy HTML code, but that is the limit. They do not—and cannot—go deeper. The openness of HTML allows for certain challenges to ideologies of control and commerce. Yet the increasingly closed, proprietary, and inaccessible expression of lower level codes does not permit even such limited challenge. Hegemony reasserts itself here, allowing some changes to the system but not fundamentally altering it enough to truly shift relations of power.

Fan/borg activities thus present some challenges to the hegemonic system of copyright and media control of images, but ultimately deeper levels of control remain untouched and unremarked on. Perhaps even more disturbing, even at surface level, the construction of Web pages by casual users is becoming marginalized. “Use” of the Internet is becoming a largely passive activity of browsing, chatting, and shopping. Creation is increasingly left to the “professionals.” This leads to ideologies of control tightening. And that situation will only intensify as concentration of ownership continues and the “grid of control” increases.

Technology is not neutral—code is redefining use, and users are being coded out of participation. Cyborg culture represents a profound shift from print culture to the information culture of control. Print culture unlocked access to knowledge for many, and information culture promises to do the same. Yet the price for participating in information culture is growing steadily higher, and instead, people are being encouraged to simply be users. In that climate, cyborg culture and the use of information technologies continue to be about conservative and capitalist desires—and the transgressive potentials of the fan/borg are being coded out of existence.

Notes

1. The television show *Buffy the Vampire Slayer* was preceded by a 1992 film of the same name. The creator of both, Joss Whedon, has said that he wanted to create a story about a female character who would be able to walk down dark alleys and defend herself against whatever might follow. The resulting television series, which played first on the WB Network and is currently on UPN, has proven popular with teens and adults. The show is a mixture of horror, drama, and comedy, commenting ironically on teenage life and life in general. The show has spawned a spin-off, *Angel*, which features the vampire of the title. Angel is a vampire with a soul (and is the ex-boyfriend of Buffy) who has headed to Los Angeles to fight crime and his own inner demons. Both shows continue to do well and have been renewed through 2003.

2. Fans of *Star Trek*, as is well known, saved the initial series from cancellation with a write-in protest started by Bjo Trimble. Although the initial series lasted only three seasons, continuing fan interest and the rise of “cons” or conventions to celebrate “all things *Star Trek*” led to the

creation of the motion picture series (nine to date and one in the works) and the development of new series—*The Next Generation* (1987-94), *Deep Space Nine* (1993-99), *Voyager* (1995-2001), and the latest series, *Enterprise* (2001-present).

3. WYSIWYG stands for “what you see is what you get,” meaning that when creating a Web page in a particular software program, you allegedly see the page as it will look online. Most software now does that, allowing the user to see how fonts actually will look when printed out and see special formatting, such as bold and italics, almost exactly as they will appear on the printed (or electronic) page.

4. It is interesting to note (and other researchers have mentioned this too) that for many Internet users, the World Wide Web is the Internet, not simply one application of it. This is especially true for Internet users that began using the Net after graphical Web browsers became popular and surfing the Web began to displace other Internet activities such as Usenet newsgroups, listservs, and FTP.

5. Ullman (1995) wrote that computer engineers, who are mostly male, “behave like children,” which is generally accepted as idiosyncratic behavior (p. 138). Furthermore, their way of living and working is best described as “alone, out-of-time, disdainful of anyone far from the machine” (p. 143). Ullman suggested that these ways of working and living are being programmed into computers, and she questioned whether that is a lifestyle most people would like to lead.

6. Web rings can be started and maintained by anyone with the interest and some basic technical skills. However, advertisers have already seen the potential for profits in Web rings, and WebRing, a free service for those interested in finding or organizing Web rings, has appeared. The site has actually been in existence since 1995 and claims to manage “tens of thousands of rings.” It makes the claim to advertisers that “every Ring in WebRing represents an affinity group (a demographic group or interest area) that in turn represents some of the most targeted advertising possibilities on the Internet today” (WebRing Advertising Department 3 August 2000, <http://www.webring.com>).

7. In spring 2001, the five-year contract between Fox and the WB Network expired, and the WB refused to pay the higher per-episode price that Fox then demanded for renewal. In response, Fox took the unusual step of moving *Buffy* to UPN, which did agree to a higher price—reportedly two million dollars per episode (Marsh 2001).

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Mia Consalvo is an assistant professor in the School of Telecommunications at Ohio University. Her research investigates uses of popular culture in new media, including the Internet and video games. She has published articles in Television & New Media and Feminist Media Studies and has most recently coedited (with Susanna Paasonen) the volume Women and Everyday Uses of the Internet: Agency and Identity for Peter Lang.